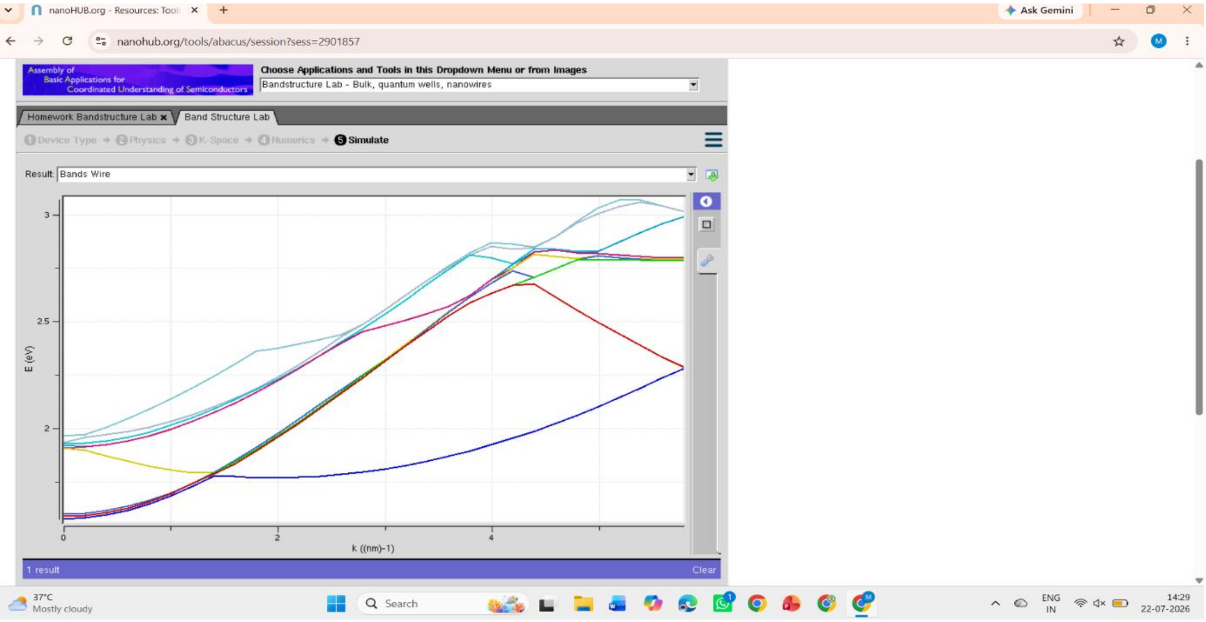
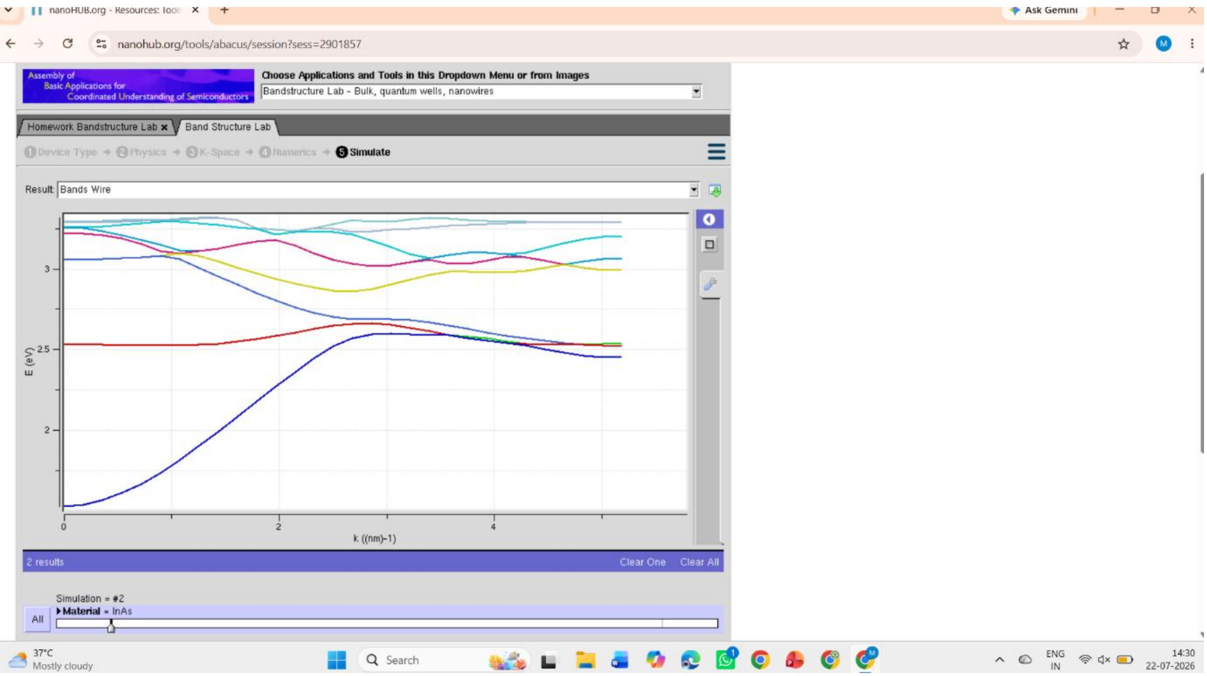
Exp. 5. Band Structure Simulation for Graphene and CNTs using Nano Hub

Case Study 1: Simulate the band-structure of InAs circular nanowire



The graph shows the **energy band structure (E vs k)** of the semiconductor, where the energy of electrons changes with the wave vector. The different colored curves represent different energy bands, indicating the allowed energy levels for electron movement.

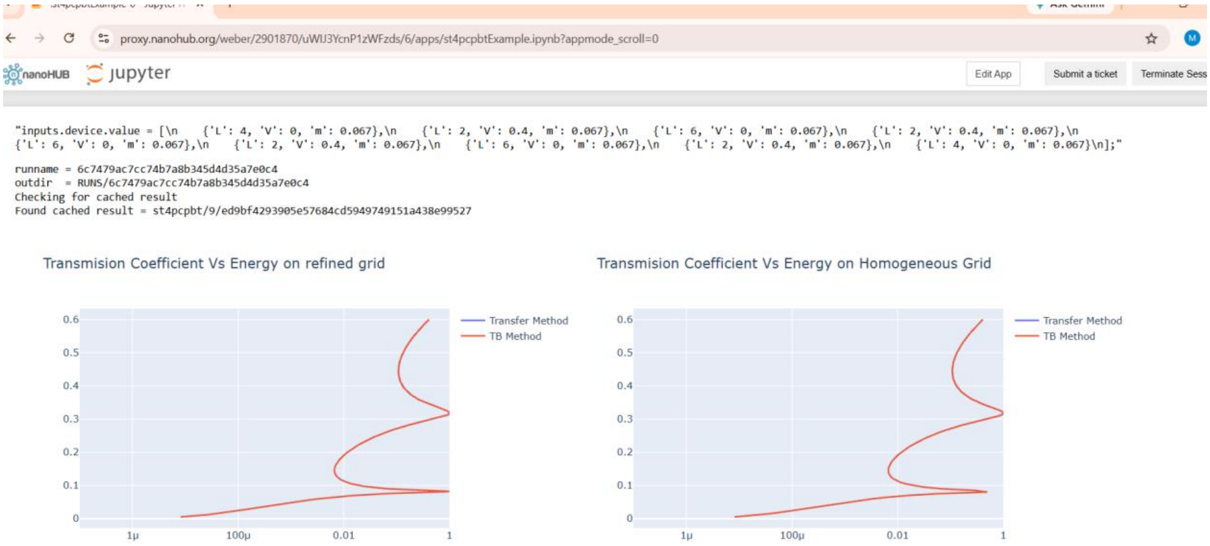
Case Study 2: Simulate the band-structure of AlGaAs circular nanowire



The graph compares **two band structure simulation results**, showing how the energy bands change with different material or simulation parameters. The variation in the curves indicates changes in the electronic properties, while the overall band structure remains consistent.

EXP. 6. Demonstration of Quantum Tunneling

Case study 1 : Analysis the transmission co-efficient Vs Energy characteristics on refined grid

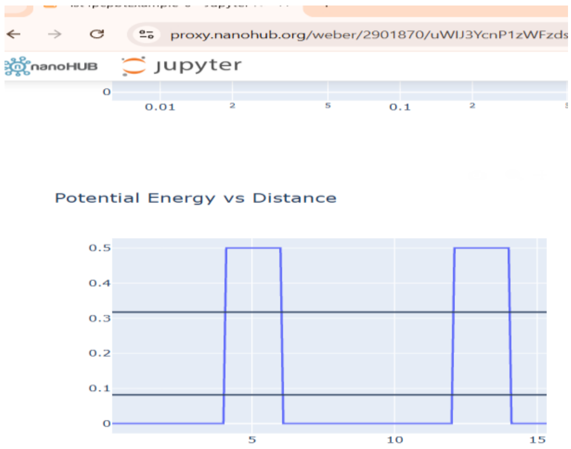


The transmission coefficient increases with electron energy, showing that electrons have a higher probability of passing through the potential barriers at higher energies. Both the Transfer Matrix and Tight-Binding methods produce nearly identical results, confirming the accuracy of the simulation.

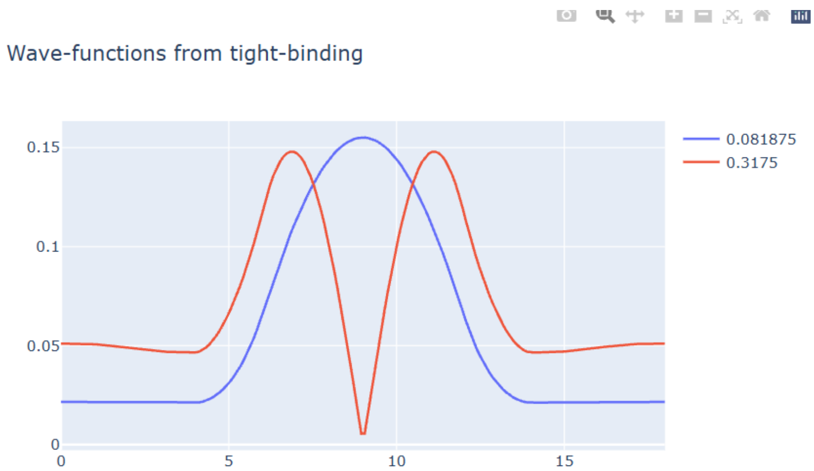
Case study 2 : Analysis the Reflection co-efficient Vs Energy on Homogeneous Grid

The reflection coefficient decreases as the transmission coefficient increases, indicating that fewer electrons are reflected at higher energies. The close overlap of both methods validates the consistency and reliability of the obtained results.

OUTPUT:



The graph shows a double potential barrier structure where the barrier height is 0.5 eV and the regions between barriers represent quantum wells. This potential profile controls the movement and tunneling behavior of electrons through the device.

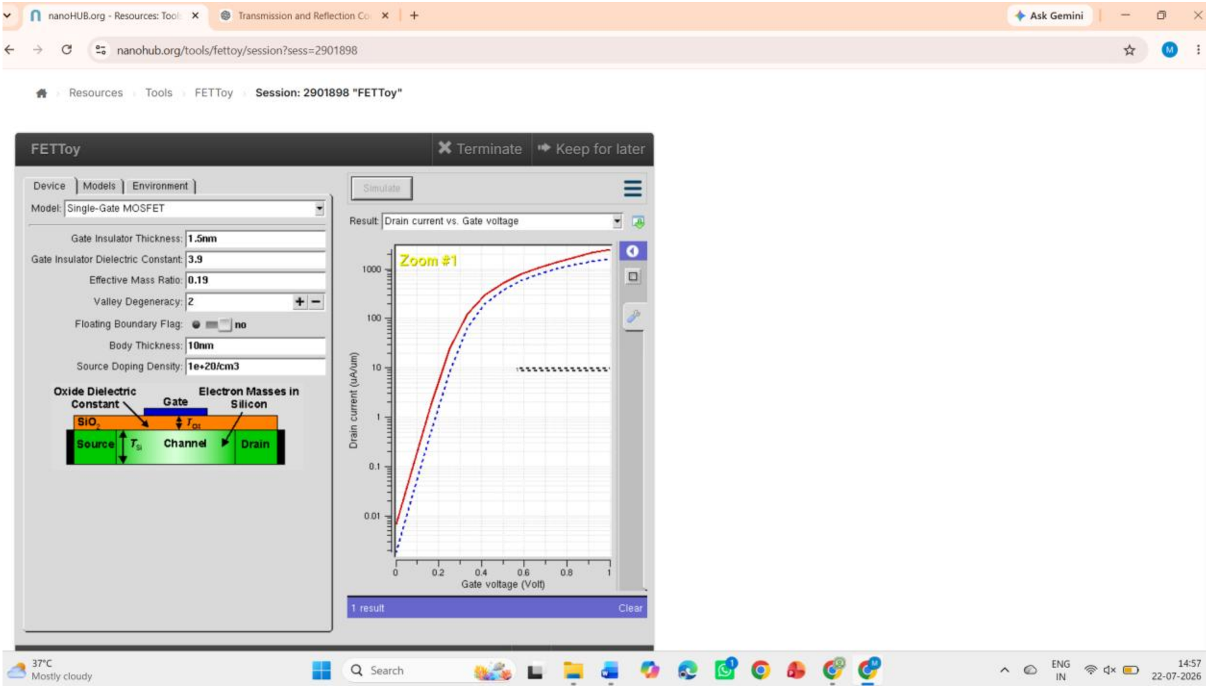


The wave functions are concentrated inside the quantum well, indicating a high probability of finding electrons in this region. The different energy states exhibit distinct wavefunction shapes, demonstrating quantum confinement and resonant behaviour.

Exp. 7. MOSFET Scaling Analysis and Subthreshold Swing Calculation using Nano Hub

Case 1: Increase in VG :

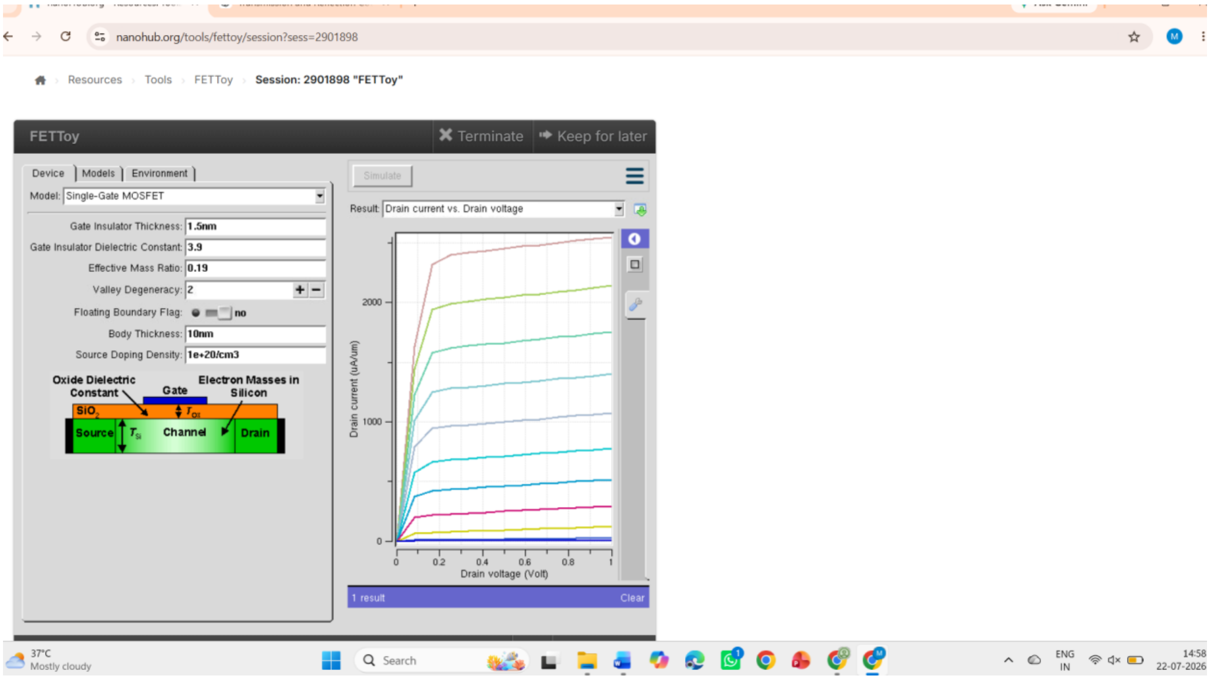
● Observation: As VG increases, the drain current (ID) also increases exponentially due to the MOSFET entering the saturation region earlier.

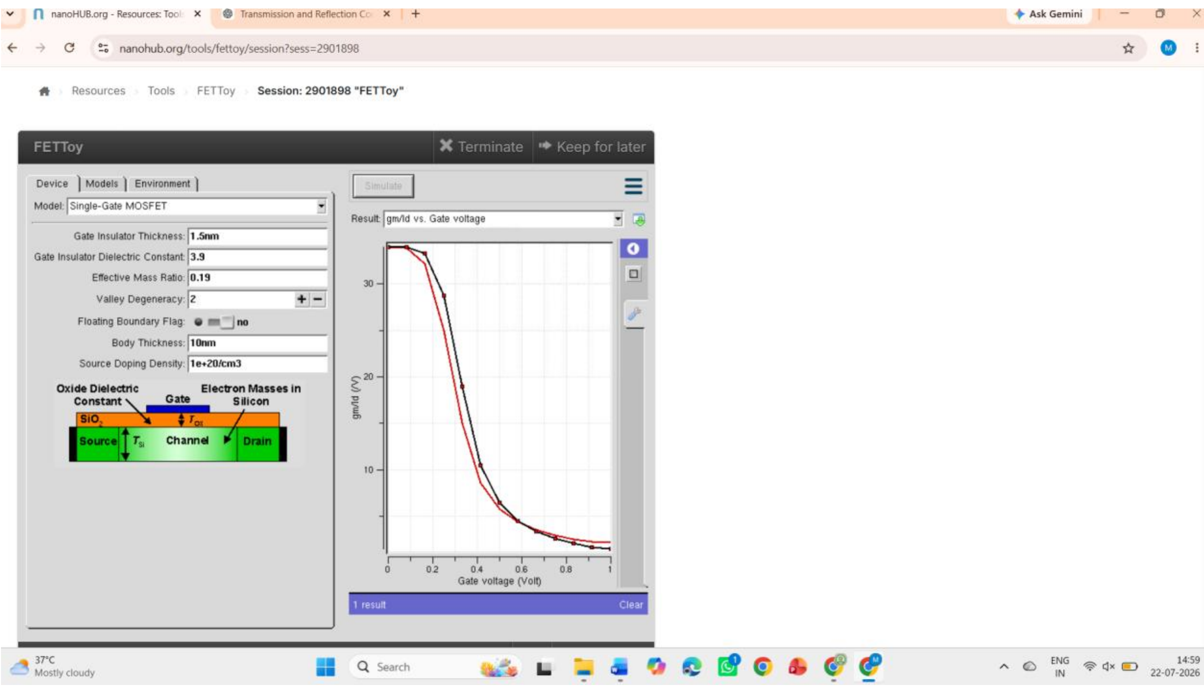


The graph shows that the drain current (Id) increases rapidly with increasing drain voltage (Vd) and then reaches a saturation region. Higher gate voltages produce higher drain currents, demonstrating the normal operating characteristics of a Single-Gate MOSFET.

Case 2: Variation in VD :

● Observation: Drain current (ID) saturates for higher VD at a fixed VG:





The graph shows that the drain current (Id) increases exponentially after the threshold voltage is reached as the gate voltage (Vg) increases. This confirms that the MOSFET turns ON and allows more current to flow with increasing gate voltage.